

CELL SIGNALING

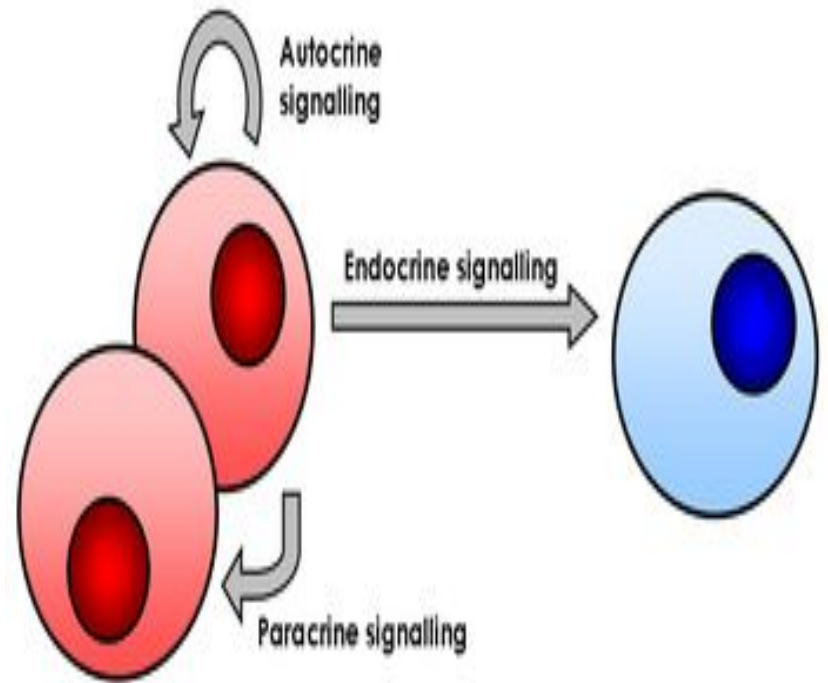
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Learning outcomes

- At the end of the lecture, student should be able to:
- Discuss the 5 types of cell signaling mechanism (contact dependent, paracrine, autocrine, endocrine, synaptic – signaling)
- Describe a second-messenger system and discuss its importance in human physiology
- Describe the roles played by cAMP, adenylate cyclase, and kinases play in cellular function

Introduction

- Cells of our body do not live in isolation.
- Cells send and receive information (signals) to initiate a response
- They communicate with one another by liberating signal molecules e.g., Hormones and neurotransmitters to control vital processes including metabolism, movement, secretion and growth.



Cell Signaling

- The process by which a cell responds to substances outside the cell through signaling molecules found on the surface of & inside the cell.
- The resulting cell signaling pathway provides amplification of the primary signal & distribution of the signal to appropriate targets within the cell.
- The activation of the transcription & subsequent translation is a common outcome of cellular signaling.

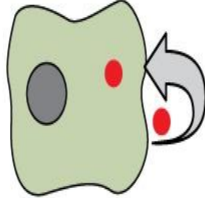
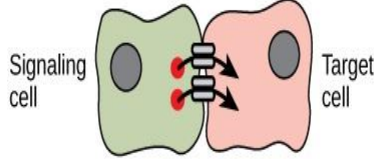
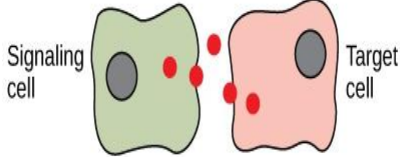
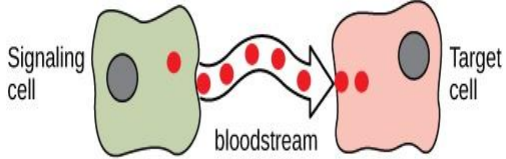
- Cell signaling is part of a complex system of communication that governs basic cellular activities and coordinates cell actions.
- The ability of cells to perceive and correctly respond to their microenvironment is the basis of development, tissue repair, and immunity as well as normal tissue homeostasis.
- Errors in cellular information processing are responsible for diseases such as cancer, autoimmunity, and diabetes.
- By understanding cell signaling, diseases may be treated effectively and, theoretically, artificial tissues may be created.

Definitions

- **Signaling:** Cell-cell communication via signals.
- **Signal transduction:** Process of converting extracellular signals into intra-cellular responses.
- **Ligand:** The signaling molecule.
- **Receptors:** Bind specific ligands and in turn activate one or more intracellular pathways. These pathways depend on intracellular signaling proteins which process the signal and transmit the signal to appropriate intracellular targets. The targets at the end of signaling pathways are called effector proteins.

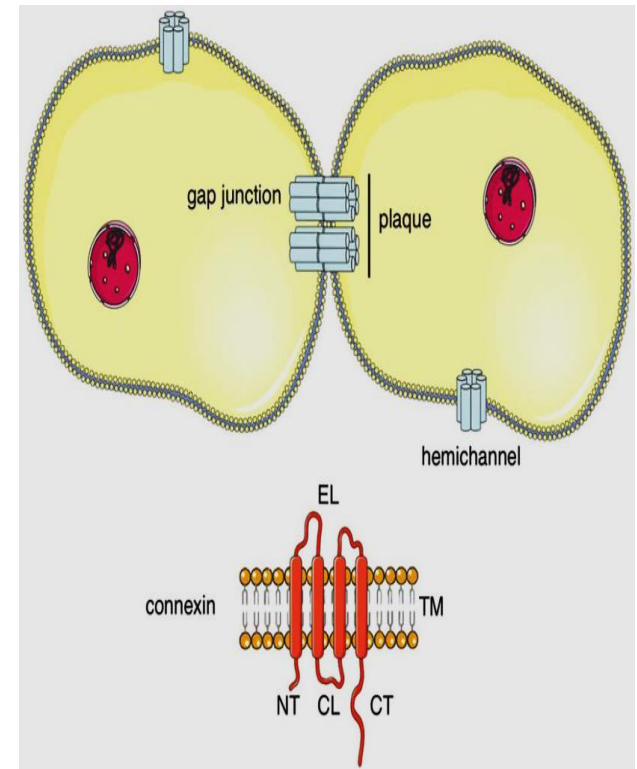
Multiple levels of Intercellular communication

- There are 5 ways by which cells communicate with one another:
- 1. Direct contact (gap junctions)
- 2. Autocrine
- 3. Paracrine
- 4. Endocrine
- 5. Synaptic

Forms of Chemical Signaling	
Autocrine	A cell targets itself.
	
Signaling across gap junctions	A cell targets a cell connected by gap junctions.
	
Paracrine	A cell targets a nearby cell.
	
Endocrine	A cell targets a distant cell through the bloodstream.
	

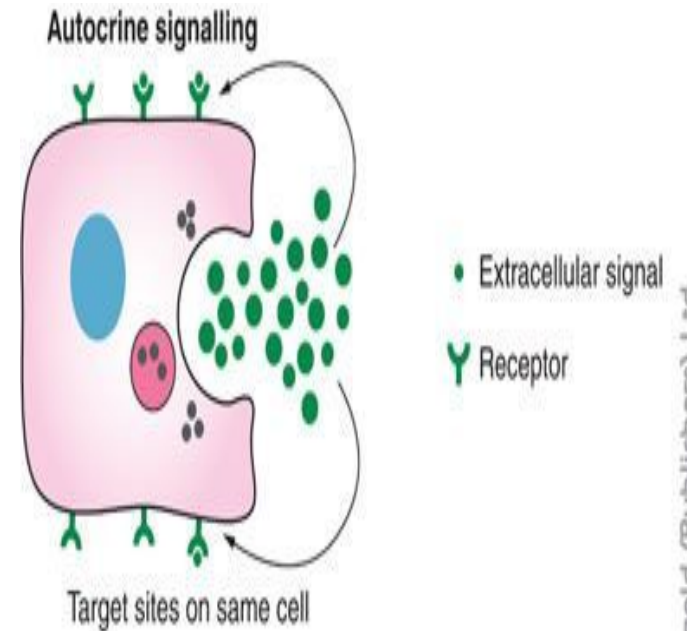
Direct contact (Gap junctions)

- Gap junctions are channels that physically connect adjacent cells.
- They mediate a rapid exchange of small molecules and ions.
- Found in cardiac muscle, smooth muscle, epithelia covering body surface etc



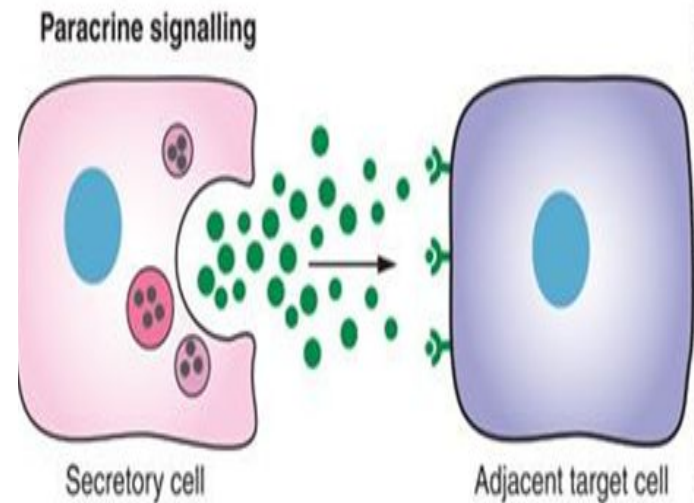
Autocrine

- It is a form of cell communication in which a cell send message to itself.
- The cell secretes a chemical substance or messenger into extracellular fluid from there it binds to autocrine receptor on the same cell leading to changes in the cell.
- For example cytokine interleukin-1 secreted from monocyte acts on it.



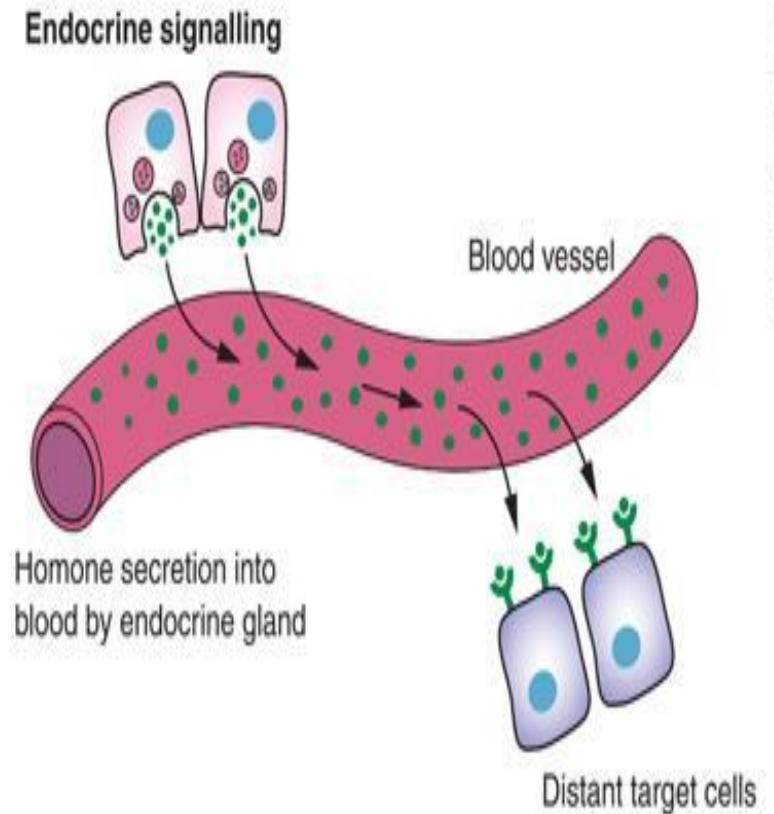
Paracrine

- It is a form of cell communication in which a cell sends a message to nearby cells to induce changes in them.
- An example is **somatostatin** secreted from **delta cells of pancreas** inhibit insulin and glucagon secretion from beta and alpha cells.



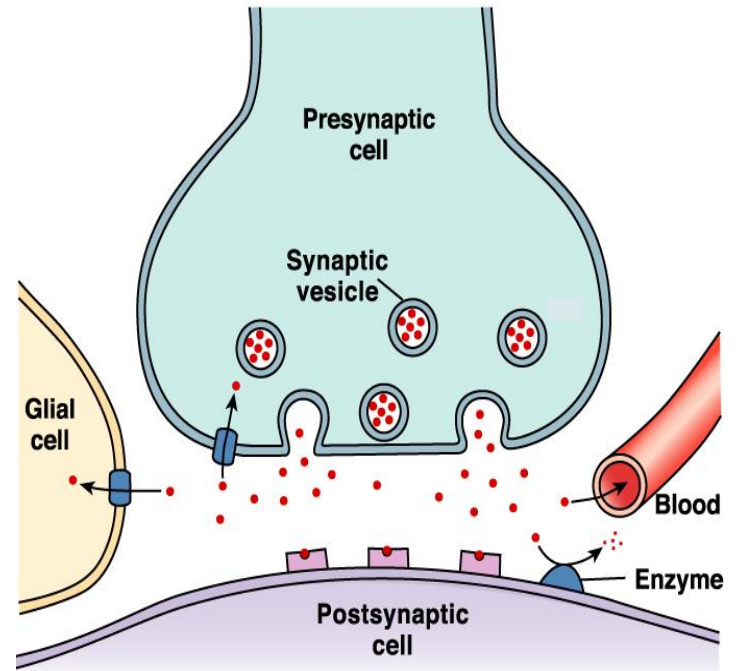
Endocrine /neuroendocrine

- Endocrine glands release hormones which are carried by blood to the target cells in distant organs.
- The hormones bind to their specific receptors present on target cells to exert their functions e.g., pituitary, thyroid hormone etc
- A neurohormone is one that is secreted from axon terminal of a neuron into blood stream instead of synapse.



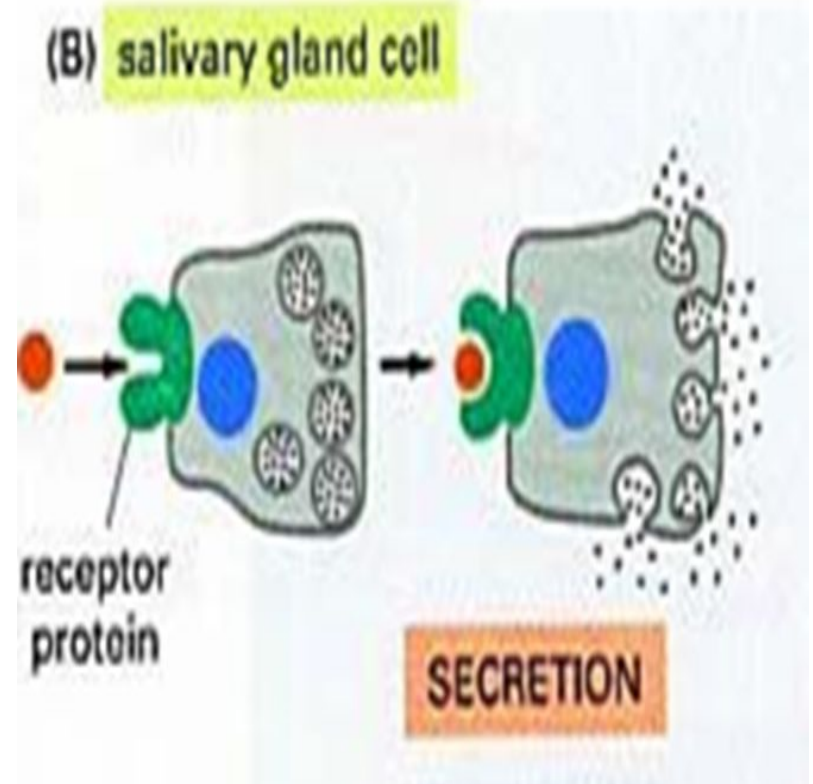
Synaptic communication

- In a neuron, synaptic vesicles store various neurotransmitters that are released at the synapse.
- The neurotransmitters then bind to receptors present on other neurons and signal is conveyed.



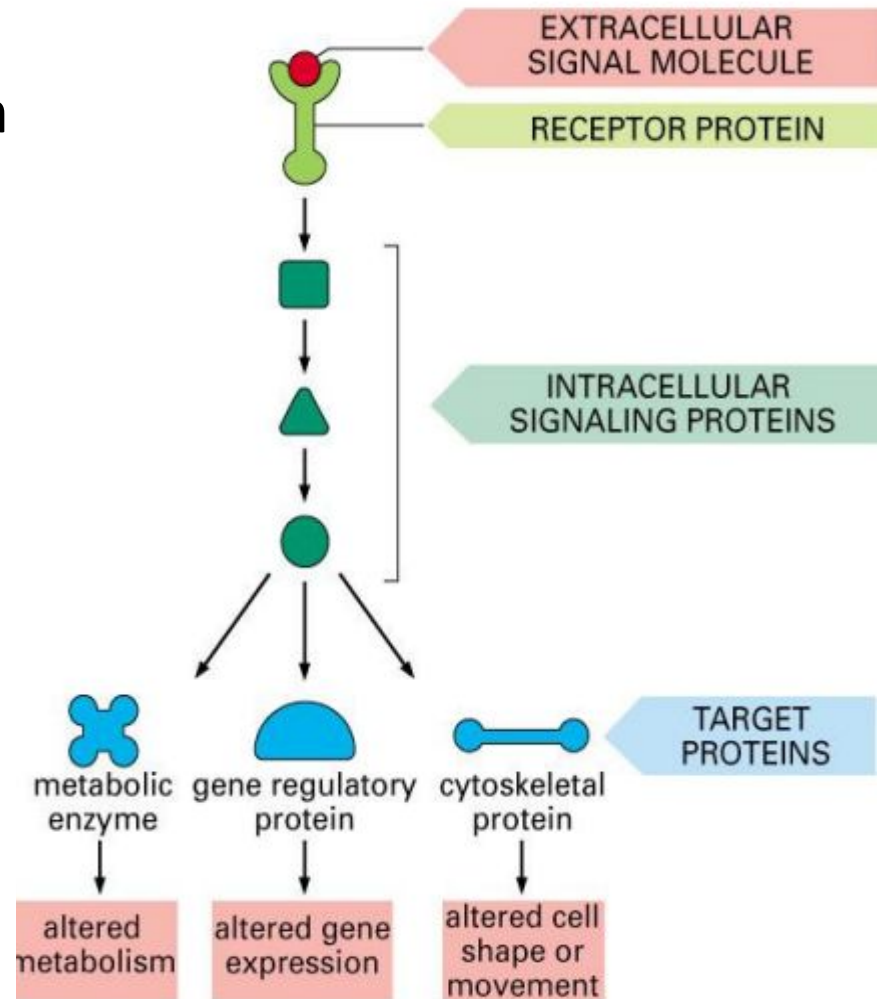
Cell to cell Communication

- Communication between cells requires:
 - Ligand: the signaling molecule, or the first messenger
 - It can be a hormone, cytokine, neurotransmitter or any chemical substance.
 - Receptor protein: molecule to which ligand binds
 - may be on the plasma membrane or within the cell
 - When ligand binds to receptor, intracellular mediators are produced called "second messengers"
 - When ligand binds to receptor, cell has a response: which results from a series of events called signal transduction.
 - Transduction is the series of events resulting in formation of 2nd messenger

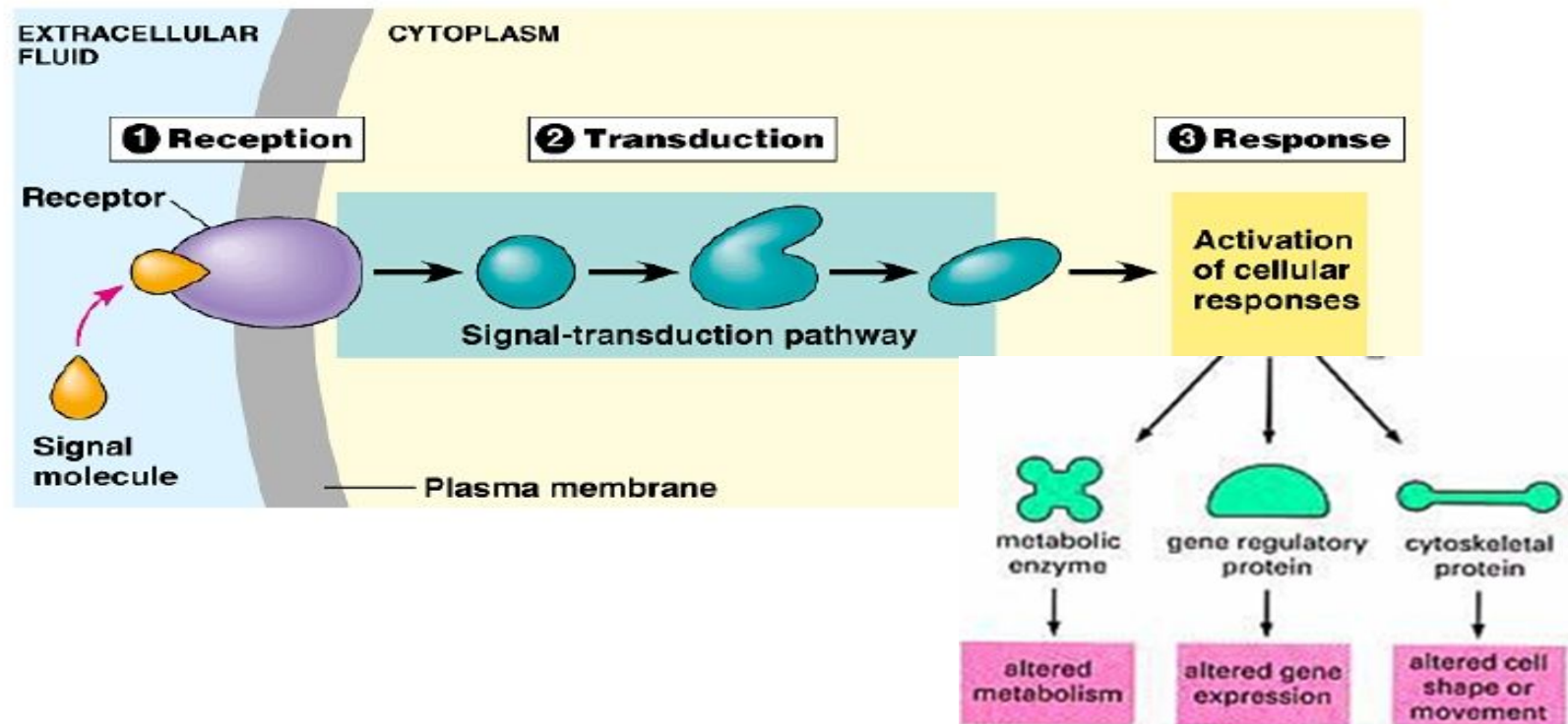


A simple intracellular signaling pathway activated by an extracellular signal molecule

- The signal molecule usually binds to a receptor protein in the PM of the target cell.
- The receptor activates one or more intracellular signaling pathways, involving a series of signaling proteins.
- Finally, one or more of the intracellular signaling proteins alter the activity of effector proteins and thereby the behavior of the cell.

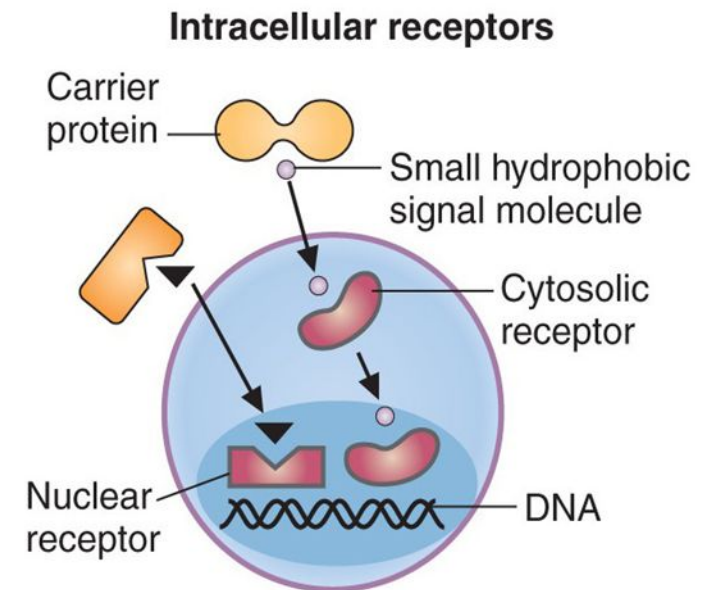
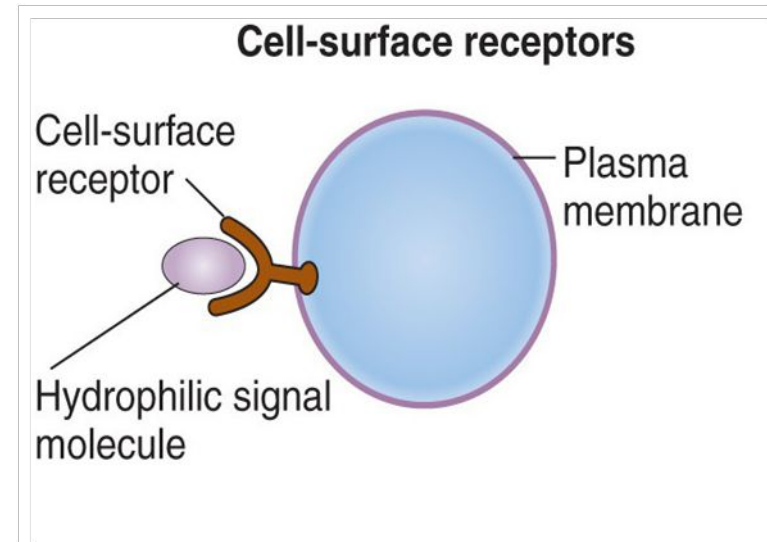


Three Stages of Signal Transduction



Types of Receptors

- Most chemical messengers are charged so can not cross the cell membrane, thus recognition is done by cell surface receptors; whereas lipid soluble substances can cross the cell membrane and bind intracellular receptor.
- Thus there are two main types of receptors
- 1. Cell membrane surface receptors
- 2. Intracellular receptors in cytoplasm or nucleus



LOCATION OF HORMONE RECEPTORS

The locations for the different types of hormone receptors are generally the following:

□ on the surface of the cell membrane.

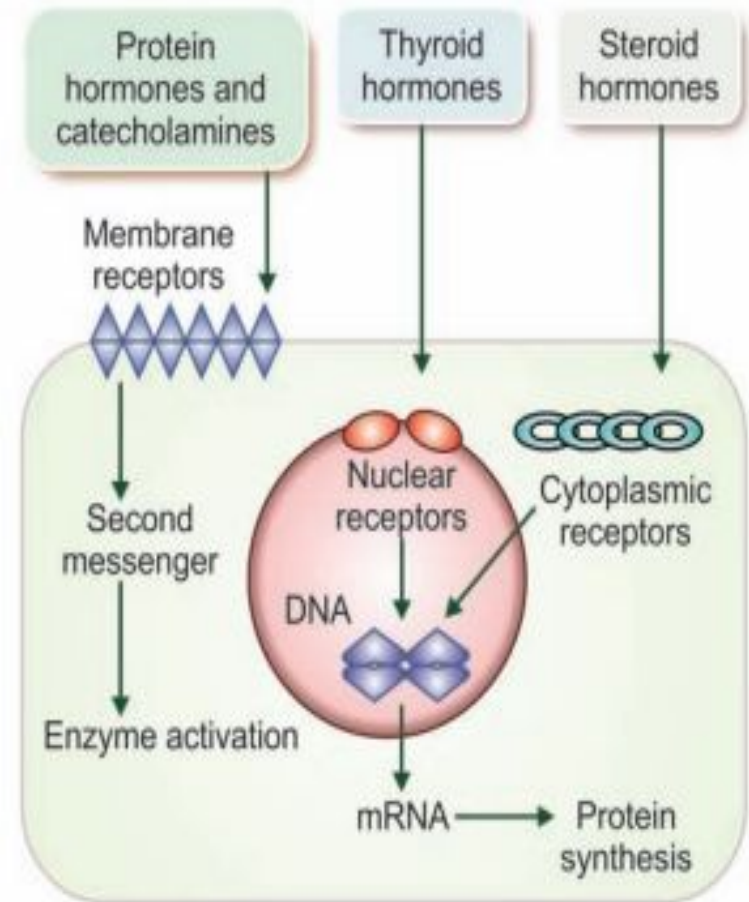
- for the protein, peptide, and catecholamine hormones.

□ In the cell cytoplasm.

- for the different steroid hormones

□ In the cell nucleus.

- for the thyroid hormones



RECEPTORS FOR CHEMICAL MESSENGERS

- The recognition of chemical messengers by cells typically begins by interaction with a receptor at that cell. There have been over 20 families of receptors for chemical messengers characterized.
- These proteins are not static components of the cell, but their numbers increase and decrease in response to various stimuli, and their properties change with changes in physiologic conditions.
- When a hormone or neurotransmitter is present in excess, the number of active receptors generally decreases (**downregulation**), whereas in the presence of a deficiency of the chemical messenger, there is an increase in the number of active receptors (**upregulation**). In its actions on the adrenal cortex, angiotensin II is an exception. it increases rather than decreases the number of its receptors in the adrenal.

4 main types of receptors for cell signaling mechanisms

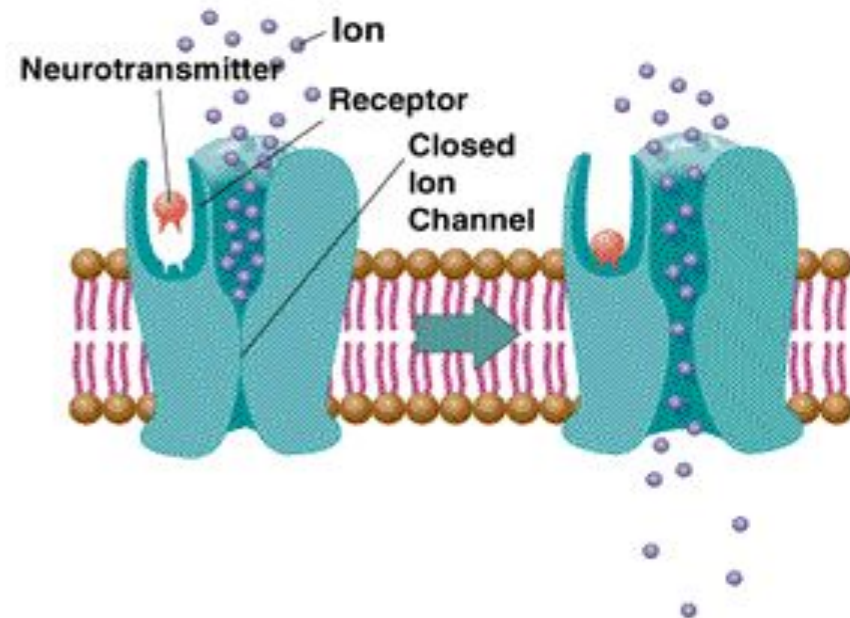
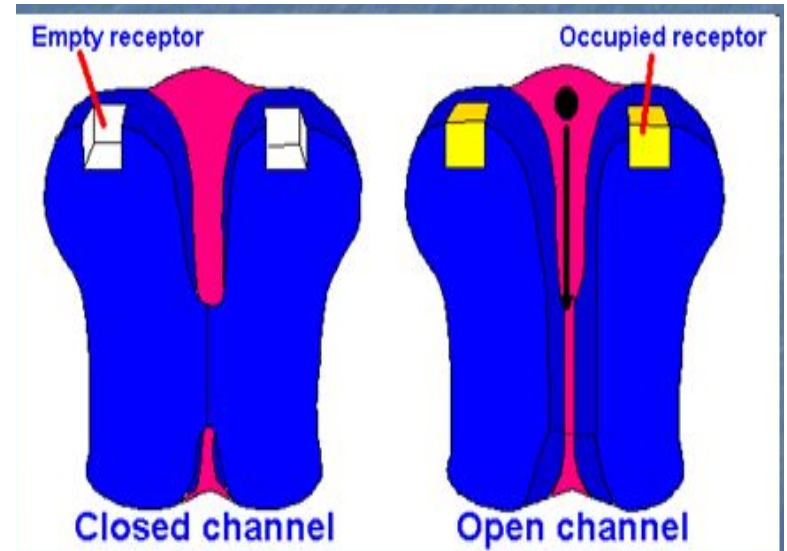
1. Ion channel linked receptor
2. G-protein linked receptor
3. Enzyme linked receptor
4. Gene activation linked receptor

Above three receptors are more common as compared to gene activation linked receptor.

Ion Channel–Linked Receptors

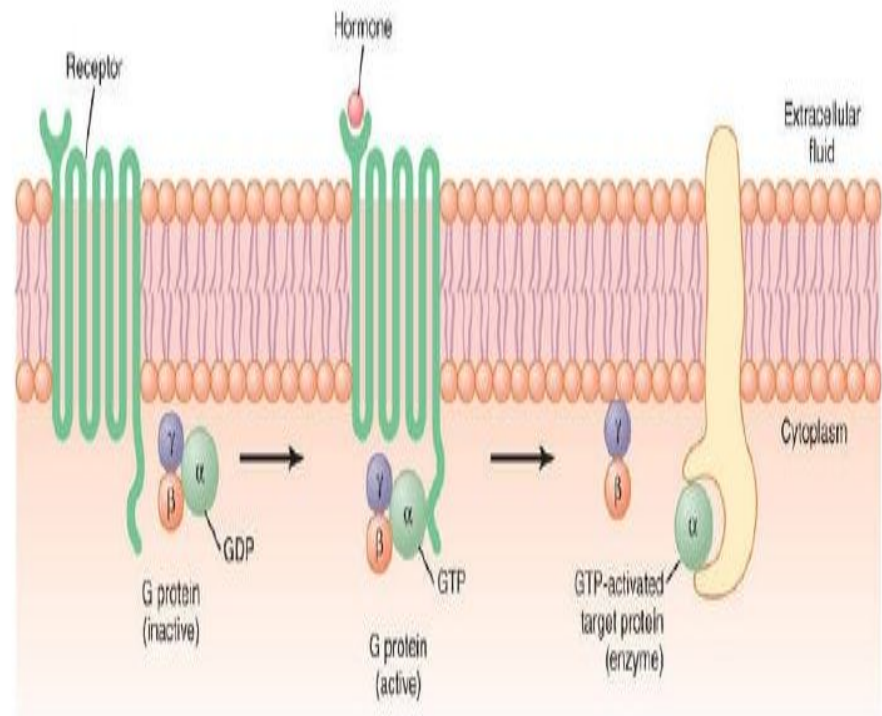
(mostly neurotransmitter)

- The ligand binds to the cell membrane receptor and ligand-receptor complex opens a channel in the cell membrane.
- It allows different electrolytes e.g., sodium, potassium to pass through channel and these ions perform a specific function



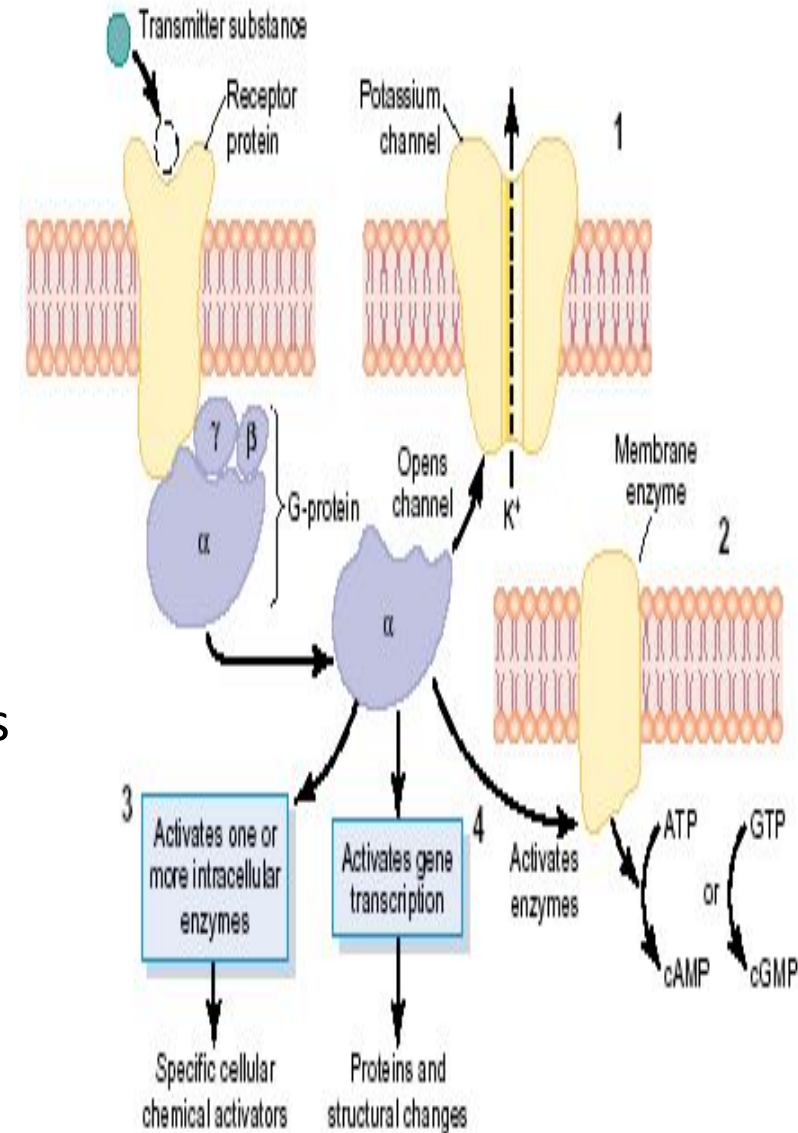
G Protein–Linked Hormone Receptors (mostly peptides and catecholamines)

- Ligand binds to the cell membrane receptor and ligand-receptor complex activates a membrane G protein attached to inner part of receptor.
- The G protein has 3 subunits: alpha, beta and gamma. They are guanylyl nucleotide (GTP/GDP) binding proteins.
- At rest, the alpha unit is bound to guanine diphosphate (GDP). It has an inherent GTPase activity thus when activated GDP is replaced by GTP.
- Binding of GTP allows G protein to dissociate from receptor and splits into activated alpha and beta-gamma subunits which are free to move.



G-protein cont'd

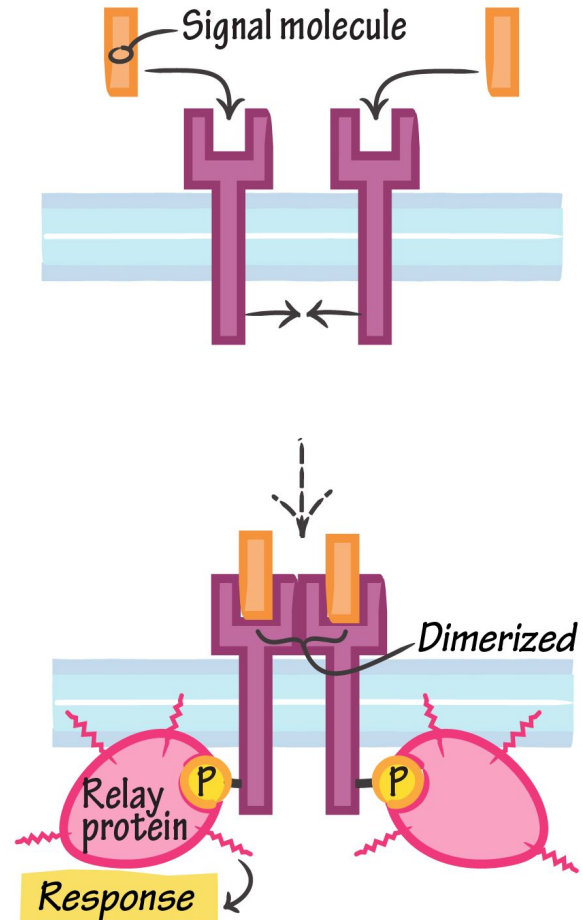
- The alpha unit mostly performs one of the following functions:
- 1. Opening of an ion channel for a longer period
- 2. Activation of a membrane enzyme
- 3. Activation of an intracellular enzyme
- 4. Gene transcription to form new proteins.
- There are several types of G proteins; G_s is stimulatory and activates adenylyl cyclase; G_i is inhibitory to adenylyl cyclase.
- G_q leads to activation of phospholipase C and generation of IP3 and DAG.



Enzyme linked receptor

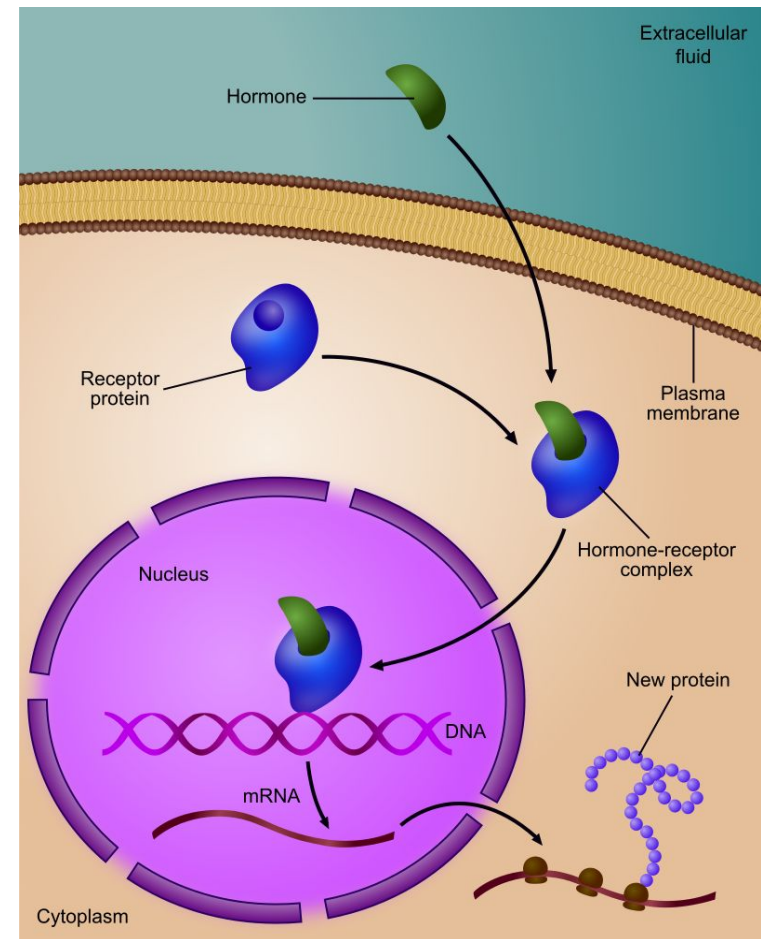
- Some receptors directly activate some enzymes attached to their inner side.
- When a ligand binds the extracellular surface of receptor, there is a conformational change in receptor and an enzyme is activated immediately inside the membrane.
- The most common example is insulin hormone that binds to tyrosine kinase receptor; active tyrosine kinase performs different cell functions.

Receptor Tyrosine Kinase



Gene activation linked receptor

- Several important hormones bind with protein receptors present inside the cells
- The activated hormone-receptor complex binds with a specific regulatory sequence of DNA called the hormone response element and activates transcription of specific genes and formation of messenger RNA.
- Newly formed proteins appear in the cell to control its function



Examples of 2nd messengers

- Commonly used 2nd messengers include:
- 1. Cyclic adenosine monophosphate (cyclic AMP)
- 2. Cyclic guanosine monophosphate (cyclic GMP)
- 3. Calcium ions
- 4. Inositol triphosphate (IP3)
- 5. Diacylglycerol (DAG)
- 6. Nitric oxide (NO)

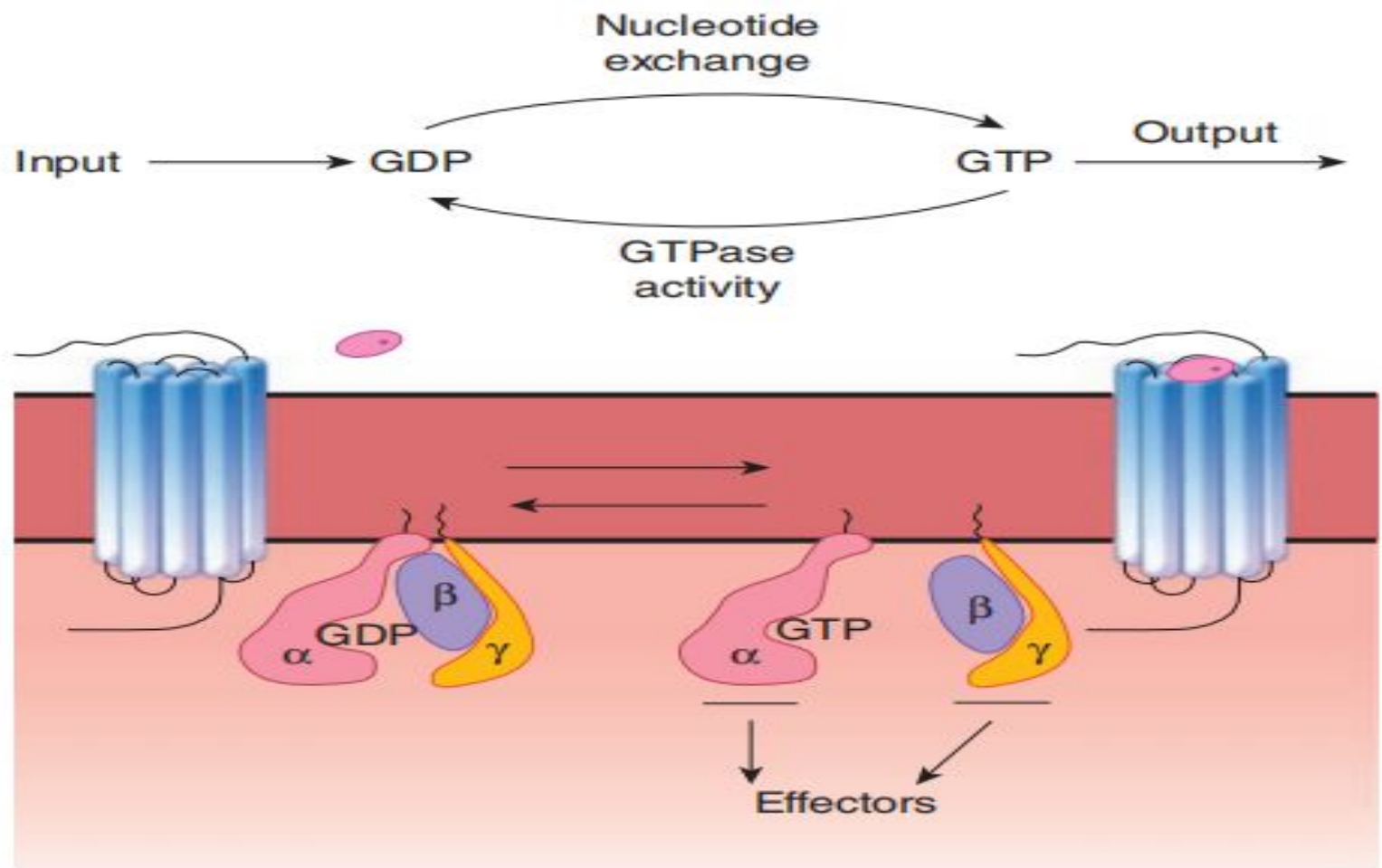


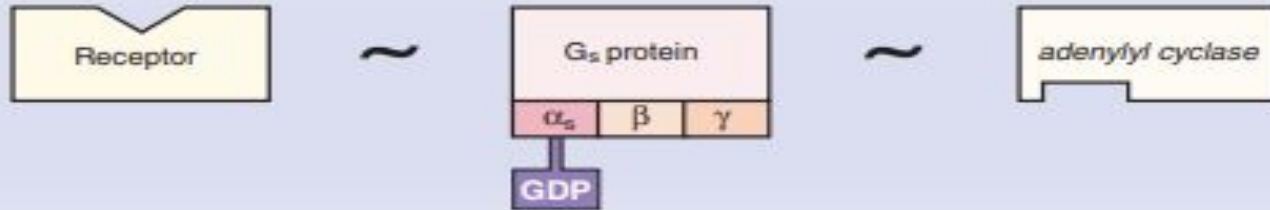
FIGURE 2-23 Heterotrimeric G-proteins. **Top:** Summary of overall reaction that occurs in the $G\alpha$ subunit. **Bottom:** When the ligand (red oval) binds to the G-protein-coupled receptor in the cell membrane, GTP replaces GDP on the α subunit. GTP- α separates from the $\beta\gamma$ subunit and GTP- α and $\beta\gamma$ both activate various effectors, producing physiologic effects. The intrinsic GTPase activity of GTP- α then converts GTP to GDP, and the α , β , and γ subunits reassociate.

cAMP signaling pathway

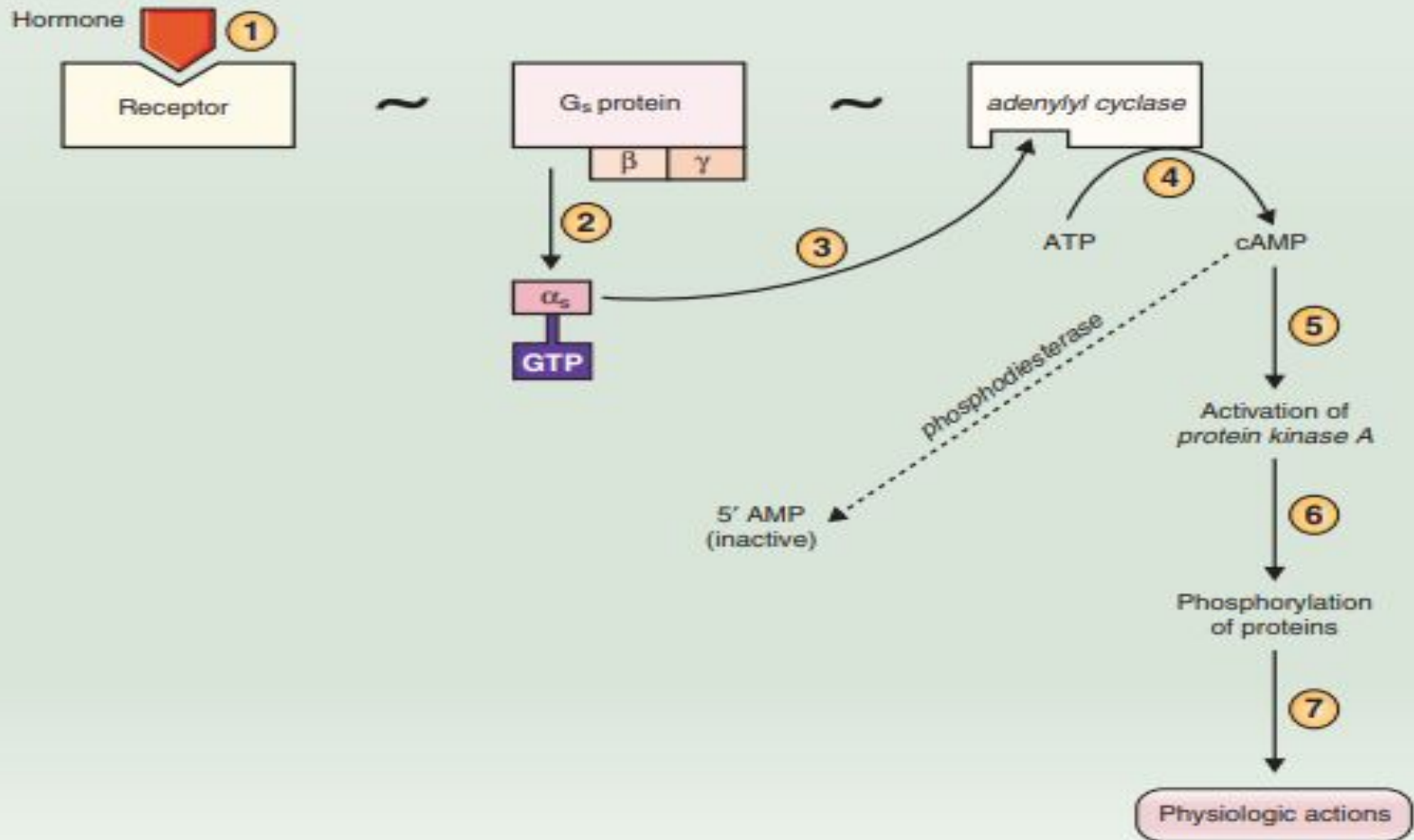
- Most of hormones use cAMP as second messenger.
- G protein activates Adenylyl cyclase enzyme which in turn causes conversion of ATP into cyclic AMP the 2nd messenger
- cAMP then carry the signal into cytoplasm where they activate protein kinase A which phosphorylates and modifies the functions of a variety of intracellular proteins.
- The effect of adenylyl cyclase is inhibited by phosphodiesterase that converts cAMP into 5'-AMP.
- Note: PKA phosphorylates serine and threonine residues

ADENYLYL CYCLASE MECHANISM

Inactive



Active

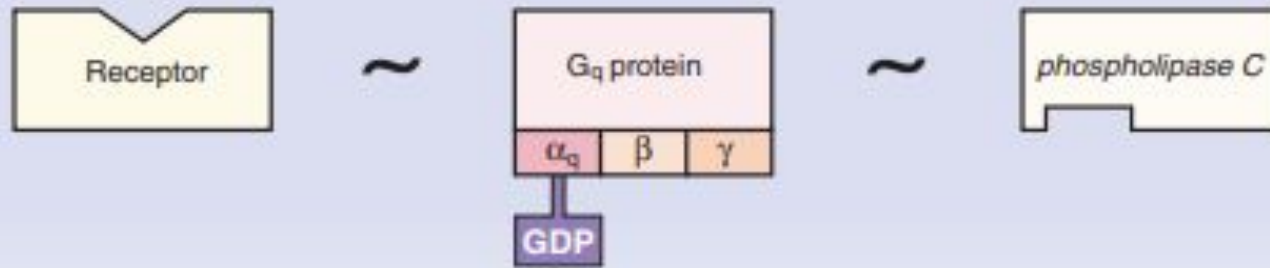


IP3 signal pathway using phospholipase C

- Activation of membrane enzyme phospholipase C (PLC) by G protein causes breakdown of phosphatidylinositol biphosphate (PIP₂) a plasma membrane lipid, into three different 2nd messengers namely inositol triphosphate (IP₃), diacylglycerol (DAG) and Calcium.
- DAG activates protein kinase C
- IP₃ diffuses into endoplasmic reticulum and triggers release of calcium into cytoplasm by binding to IP₃ receptor a ligand gated calcium channel.
- Ca⁺⁺ is an important 2nd messenger that triggers secretion, contraction and enzyme activation in different cells.
- PKC can act as a kinase to phosphorylate histones, serine and/or threonine residues

PHOSPHOLIPASE C MECHANISM

Inactive



Active

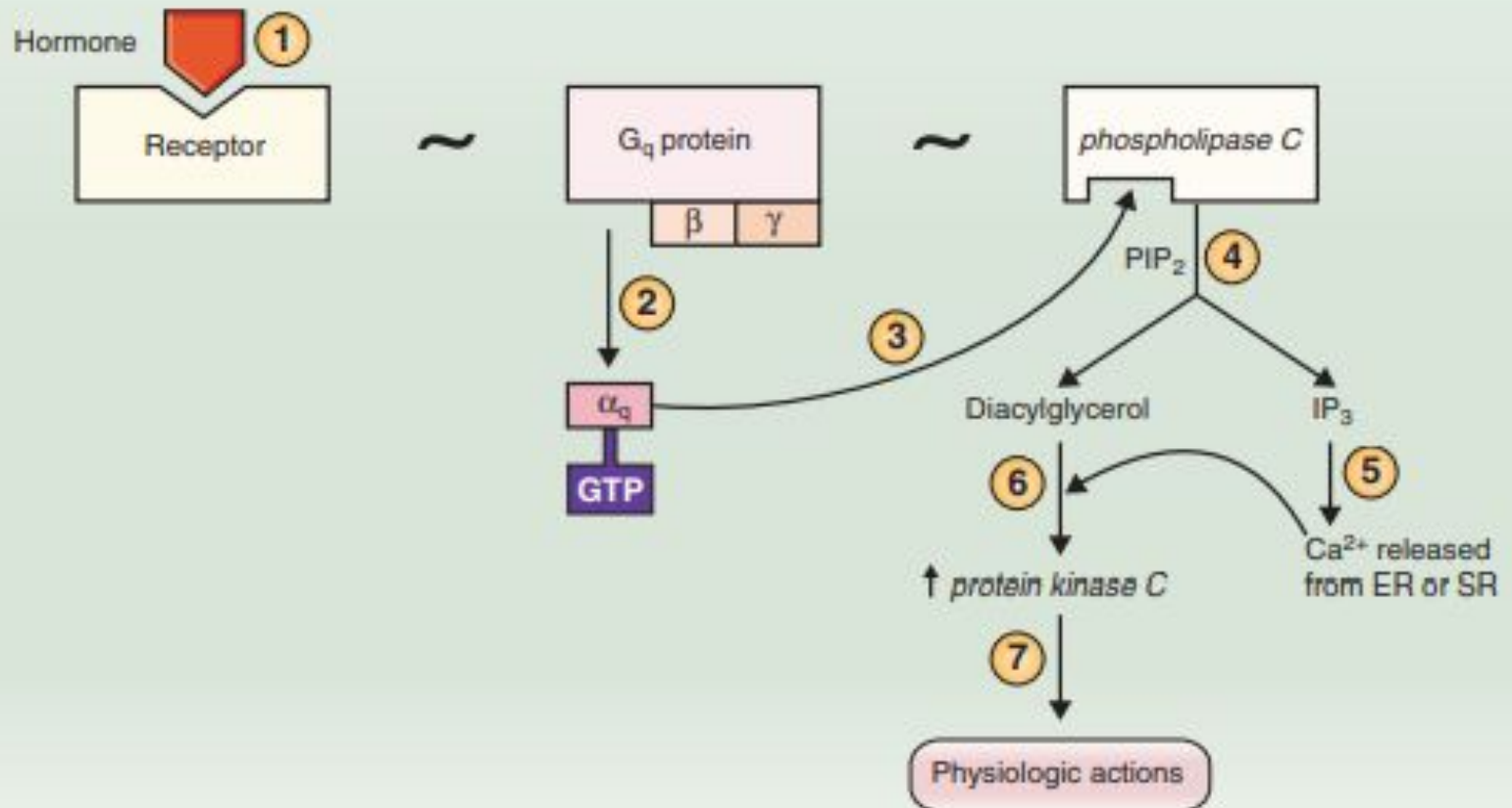


TABLE 2-3 Common mechanisms by which chemical messengers in the ECF bring about changes in cell function.

Mechanism	Examples
Open or close ion channels in cell membrane	Acetylcholine on nicotinic cholinergic receptor; norepinephrine on K ⁺ channel in the heart
Act via cytoplasmic or nuclear receptors to increase transcription of selected mRNAs	Thyroid hormones, retinoic acid, steroid hormones
Activate phospholipase C with intracellular production of DAG, IP ₃ , and other inositol phosphates	Angiotensin II, norepinephrine via α_1 -adrenergic receptor, vasopressin via V ₁ receptor
Activate or inhibit adenylyl cyclase, causing increased or decreased intracellular production of cAMP	Norepinephrine via β_1 -adrenergic receptor (increased cAMP), norepinephrine via α_2 -adrenergic receptor (decreased cAMP)
Increase cGMP in cell	Atrial natriuretic peptide, nitric oxide
Increase tyrosine kinase activity of cytoplasmic portions of transmembrane receptors	Insulin, EGF, PDGF, M-CSF
Increase serine or threonine kinase activity	TGF- β , activin, inhibin

cAMP, cyclic adenosine 3',5'-monophosphate; cGMP, cyclic guanosine monophosphate; DAG, diacylglycerol; ECF, extracellular fluid; EGF, epidermal growth factor; IP₃, inositol triphosphate; M-CSF, monocyte colony-stimulating factor; PDGF, platelet-derived growth factor; TGF- β , transforming growth factor β .

Youtube Video Link

<https://youtu.be/MoHQAyMGCFw>

